

Understanding the Primary Socioeconomic Drivers of State Test Scores, Grades 3-8

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Introduction

Standardized test scores are one of the statistics available for nearly every public school district in the country, making them a common measure to assess whether our education system is serving students well. Recent reporting on a generation-long decline in reading scores, paired with proposed cuts to federal Department of Education funding, has drawn increased attention to the matter of where funding should be allocated to improve test scores and more generally educational outcomes. Before discussing funding allocation, policymakers require a clear understanding of the data itself and the potential shared characteristics among school districts with lower test scores.

Socioeconomic status is a characteristic consistently mentioned in this conversation. Built off the correlations indicating that districts with higher incidents of poverty, and less educational attainment among adults tend to have lower-performing school test scores. But socioeconomic status bundles together several distinct conditions: income, education, employment, family structure, and use of public assistance (such as SNAP benefits utilization, etc.). Understanding and quantifying the relationship between socioeconomic status indicators and district state test scores with increased specificity is crucial when attempting to improve outcomes. A policymaker deciding how to allocate funding should understand not just that socioeconomic status is important, but which dimensions of it are most strongly associated with test performance, and how large those associations are in practice.

We approach this question as data consultants for a coalition of school boards seeking to better understand the primary drivers of achievement gaps in their districts. Our research question is purely descriptive: what is the relationship between district-level socioeconomic characteristics and average student test performance across U.S. school districts? We do not claim that any of these characteristics cause test scores to rise or fall, only that describing the strength and shape of these relationships is a necessary first step before anyone can design a targeted, evidence-based funding policy or a causal study of one.

Data

Source and Provenance

We use the Stanford Education Data Archive (SEDA), version 6.0 (Reardon et al., Stanford University), a publicly available dataset built from over 480 million standardized test records submitted by U.S. states under federal reporting requirements. SEDA converts each state's own assessments onto a common scale by linking them to the National Assessment of Educational Progress (NAEP), producing test scores that are comparable across states and years. We combine two SEDA files, both at the school-district level:

- **seda_geodist_annual_cs_6.0** (achievement): average math and reading/language-arts performance for grades 3-8,

- **seda_cov_geodist_annual_6.0** (covariates): district-level socioeconomic and demographic characteristics compiled from the American Community Survey and other federal sources.

For additional information regarding the methodology the SEDA utilized in collecting this data, as well as the files themselves, please visit their website at: <https://edopportunity.org/opportunity/data/downloads/#documentation-6>

Data Wrangling

From the achievement file, we keep only the **all students** aggregate (filtering on col **subcat = all**), which reports the achievement data in aggregate instead of by subgroups such as race or gender. Furthermore for both our achievement file and covariate file we will filter for **year = 2018**, as this is the most recent pre-pandemic year available to completely avoid any issues that may have resulted from COVID. Resulting in each row representing one achievement observation per district for 2018. We then join our data to the 2018 covariate file using the district identifier **sedalea**, which is common to both files. Districts that appear in only one file are dropped via an inner join, and rows missing any of our six predictor variables are also dropped.

Before dropping missing values, the 2018 achievement file has 11,139 district-level **all students** observations and the 2018 covariate file has 12,690. Additionally 324 districts (roughly ~2.5%) are missing covariate data entirely. To stay consistent with SEDA methodology and to preserve only complete data, we've opted to drop mismatches such that we primarily use the covariate file to supplement our achievement file data, and to drop NAs, resulting in a population dataset of 11044 observations, one per school district. As discussed in our previous proposal, districts are not drawn independently, districts neighboring each other likely have similar labor markets, demographics, and state funding resources which would naturally result in iid being violated. We will address this directly with a random sampling method rather than fitting on the full population with a resulting sample size of 1,000.

Operationalization

Outcome (Y): District test performance, **cs_mn_avg_ol** — a standardized effect size measuring how far a district's average test score falls from the national mean, in national standard-deviation units. It is metric and pools math and reading/language-arts performance across grades 3–8.

Predictors (X), all district-level, metric:

Variable	Column	Definition
Income	lninc50all	Log of median household income
Educational attainment	baplusall	% of adults with a bachelor's degree or higher
Poverty	povertyall	% of residents below the poverty line
Unemployment	unempall	% unemployment rate
SNAP receipt	snapall	% of households receiving SNAP benefits
Single motherhood	single_momall	% of households headed by a single mother

Data Prep

```
library(dplyr)
library(readr)
library(tidyr)

# Load raw SEDA files (checked into the repo, already subset to 2018)
seda_achieve <- read_csv("data/raw/seda_geodist_2018.csv",
  show_col_types = FALSE
)
seda_cov <- read_csv("data/raw/seda_cov_geodist_2018.csv",
  show_col_types = FALSE
)

# Achievement file: district-level "all students" estimate, 2018 only
achieve_2018 <- seda_achieve %>%
  filter(year == 2018, subcat == "all", subgroup == "all", gap == 0)

# Covariate file: 2018 socioeconomic predictors
cov_2018 <- seda_cov %>%
  filter(year == 2018) %>%
  select(
    sedalea, lninc50all, baplusall, povertyall,
    unempall, snapall, single_momall, totenrl, perwht,
    perblk, perhsp, perasn, pernam
  )

# Join achievement and covariates on district id (sedalea)
seda_population_file_2018 <- achieve_2018 %>%
  inner_join(cov_2018, by = "sedalea") %>%
  drop_na()

nrow(seda_population_file_2018)
```

Sampling Methodology

Data & Sampling

Our data comes from the 2018 wave of the Stanford Education Data Archive (SEDA), a nationally comprehensive source that provides district-level academic performance and demographic information for public school districts across the United States. We work with the all-subgroups population file, which gives us one record per district, and restrict the frame to districts with positive enrollment and complete data on all variables we intend to analyze. That cleaning step leaves us with a well-defined population of districts before we draw a single observation.

The variables we retain reflect the core question. Our outcome is each district's average test score composite. Our socioeconomic predictors — bachelor's degree attainment, poverty rate, unemployment, SNAP receipt, single-mother household share, and log median income — are kept as separate measures rather than collapsed into a single index, precisely because we want to understand which dimensions of socioeconomic status carry the most weight. We also retain total enrollment and race and ethnicity shares (white, Black, Hispanic, Asian, and Native American) as contextual controls.

From that cleaned frame we draw a simple random sample of 1,000 districts without replacement. SRS is the right choice here because it satisfies the IID assumption directly: every district is selected independently and with equal probability, with no clustering or stratification introducing dependence into the sample. A fixed random seed ensures the draw is reproducible.

We then immediately split the sample before conducting any analysis. Three hundred districts (30%) are assigned to an exploration set; the remaining 700 (70%) are held as a confirmation set. All variable selection, distributional checks, and collinearity diagnostics are carried out on the exploration set only. The confirmation set does not inform any modeling decisions — it is reserved entirely for final estimation. This discipline matters because allowing the same data to guide both specification and estimation inflates the risk of finding patterns that do not replicate. Before writing any output, five automated checks verify that the sample is correctly sized, contains no missing values on analysis variables, includes only districts with positive enrollment, has unique district identifiers throughout, and that the exploration and confirmation counts sum correctly to 1,000.

```
library(tidyverse)
set.seed(42) # answer of the universe
TARGET_N <- 1000
EXPLORE_FRAC <- 0.30
# (subgroup == "all", gap == FALSE, year 2018).
pop <- seda_population_file_2018
id_vars <- c("sedalea", "sedaleaname", "stateabb", "fips")
outcome <- "cs_mn_avg_ol"
outcome_se <- "cs_mn_avg_ol_se"
ses_comp <- c(
  "bapplusall",
  "povertyall",
  "unempall",
  "snapall",
  "single_momall",
  "lninc50all"
)
enrollment <- "totenrl"
race <- c("perwht", "perblk", "perhsp", "perasn", "pernam")
analysis_vars <- c(outcome, ses_comp, enrollment, race)
frame <- pop %>%
  select(all_of(c(id_vars, outcome_se, analysis_vars)))
n_start <- nrow(frame)
frame_clean <- frame %>%
  filter(totenrl > 0) %>%
  drop_na(all_of(analysis_vars))

n_dropped <- n_start - nrow(frame_clean)
cat(sprintf(
  "Frame: %d districts -> %d after cleaning (dropped %d, %.1f%%)\n",
  n_start, nrow(frame_clean), n_dropped, 100 * n_dropped / n_start
))
# validate its IID
stopifnot(TARGET_N <= nrow(frame_clean))
sample_df <- frame_clean %>%
  slice_sample(n = TARGET_N) %>%
  mutate(split = if_else(row_number() <= floor(EXPLORE_FRAC * TARGET_N),
    "exploration", "confirmation"
  ))
```

```

run_checks <- function(df) {
  results <- c()
  stopifnot(nrow(df) == TARGET_N)
  results <- c(results, sprintf("PASS Test 1: sample has %d rows", nrow(df)))
  stopifnot(!any(is.na(df %>% select(all_of(analysis_vars)))))
  results <- c(results, "PASS Test 2: no NAs in the analysis variables")
  stopifnot(all(df$totenrl > 0))
  results <- c(results, "PASS Test 3: every district has totenrl > 0")
  stopifnot(n_distinct(df$sedalea) == nrow(df))
  results <- c(results, "PASS Test 4: all sedalea district ids are unique")
  n_expl <- sum(df$split == "exploration")
  n_conf <- sum(df$split == "confirmation")
  stopifnot(n_expl + n_conf == nrow(df))
  results <- c(results, sprintf(
    "PASS Test 5: split adds up (exploration=%d, confirmation=%d)",
    n_expl, n_conf
  ))
  cat(paste(results, collapse = "\n"), "\n")
  cat(sprintf("\n%d/5 checks passed.\n", length(results)))
}
run_checks(sample_df)
cat(sprintf(
  "Done! All checks successfully pass for our iid sample of size = %d\n",
  TARGET_N
))

```

Model Building

All model choices below are based only on the exploration data. We do not use the confirmation set until the very end, so none of the decisions will introduce bias in the final results.

```

library(lmtest)
library(sandwich)
library(stargazer)

# Filter sample_df down to just the rows tagged "exploration" converts from a tibble to a plain data frame
explore <- as.data.frame(sample_df %>% filter(split == "exploration"))

# Quick check to show how many districts ended up in the exploration set
cat(sprintf("Exploration set: %d districts\n", nrow(explore)))

```

```
## Exploration set: 300 districts
```

We also create three unique models

Model 1 showcases a comparison between test scores vs. poverty rate only. Consider this our starting point because poverty is the measure most used in policy discussions

Model 2 will add the other five factors (income, education, employment, SNAP, family structure); This shows us how much of poverty's impacts on its own then with the other factors

Model 3 adds a squared poverty term so that we can check if the relationship between the two is a straight line or curves at high levels of poverty

We use robust standard errors instead of regular ones since this type of district-level data commonly has uneven variance.

```
# Model 1: starting point; poverty rate only with no controls
m1 <- lm(cs_mn_avg_ol ~ povertyall, data = explore)

# Model 2: full socioeconomic set that add other factors
m2 <- lm(cs_mn_avg_ol ~ povertyall + lninc50all + baplustall +
  unempall + snapall + single_momall, data = explore)

# Model 3: nonlinearity check adds squared poverty term to test whether the poverty-test score relation
m3 <- lm(cs_mn_avg_ol ~ povertyall + I(povertyall^2) + lninc50all +
  baplustall + unempall + snapall + single_momall, data = explore)

# District-level socioeconomic data often has non-constant error variance, so we compute HC (heterosked
robust_se <- function(model) sqrt(diag(vcovHC(model)))
```

This chunk builds the side-by-side regression table comparing for model 1,2,and 3.

Table 1: Regression Results (Exploration Set)

	<i>Dependent variable:</i>		
	Test Performance (SD units from national mean)		
	(1)	(2)	(3)
Poverty Rate	−4.02*** (0.29)	−1.24* (0.57)	−2.59* (1.18)
Poverty Rate Squared			4.14 (3.09)
Log Median Income		0.03 (0.12)	−0.01 (0.12)
% Bachelor's Degree+		1.30*** (0.18)	1.26*** (0.19)
Unemployment Rate		−2.30* (1.10)	−2.38* (1.11)
% SNAP Recipients		0.20 (0.48)	0.14 (0.48)
% Single-Mother Households		−1.19*** (0.36)	−1.23*** (0.35)
Constant	0.51*** (0.04)	−0.16 (1.27)	0.32 (1.32)
Observations	300	300	300
R ²	0.45	0.63	0.63
Adjusted R ²	0.45	0.62	0.62

Note:

*p<0.05; **p<0.01; ***p<0.001